Feynman Prescription, Residues, and Regularization of Real Poles

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Abstract

These notes develop the Feynman prescription starting from a model integral with poles on the real axis. We first introduce residues through the Laurent series, including simple poles and higher-order poles, and explain how infinitesimal detours around real poles produce contributions proportional to $\pm i\pi$ Res. We then compute the Cauchy principal value using deformed contours. Finally, we introduce the $+i\epsilon$ prescription as a rule that moves the poles off the real axis, obtain the regularized answer in the usual textbook notation, and connect the construction with the Feynman propagator in quantum field theory.

1 A Model Integral with Real Poles

Consider the integral

$$\boxed{\int_{-\infty}^{+\infty} dx\, \frac{e^{-iax}}{x^2 - b^2}, \qquad a \in \mathbb{R}, \qquad b > 0.} \tag{1}$$

The denominator factorizes as

$$x^2 - b^2 = (x - b)(x + b), \tag{2}$$

so the integrand has simple poles at

$$x = b, \qquad x = -b. \tag{3}$$

The difficulty is that both poles lie directly on the path of integration. Therefore the algebraic expression alone does not define the integral: one must specify how the path avoids the poles in the complex plane.

2 Passing to the Complex Plane

We extend the real variable x to a complex variable z, and consider

$$f(z) = \frac{e^{-iaz}}{z^2 - b^2} = \frac{e^{-iaz}}{(z - b)(z + b)}. \tag{4}$$

The problem is now geometric: the real line is an integration contour, and the singularities $z = \pm b$ lie on that contour.

3 Laurent Series and Residues

Definition 1 (Punctured neighborhood)**.** If $z_0 \in \mathbb{C}$, a punctured neighborhood of z_0 is a region of the form

$$0 < |z - z_0| < R. \tag{5}$$

The point z_0 itself is removed. This is precisely the natural local region around an isolated singularity.

Definition 2 (Laurent series)**.** Near an isolated singularity z_0, a function may admit an expansion

$$\boxed{f(z) = \sum_{n=-\infty}^{+\infty} c_n (z - z_0)^n = \cdots + \frac{c_{-2}}{(z - z_0)^2} + \frac{c_{-1}}{z - z_0} + c_0 + c_1(z - z_0) + \cdots.} \tag{6}$$

The terms with negative powers form the principal part; the terms with nonnegative powers form the regular part.

Example 1 (A pole with a regular part)**.** Take

$$f(z) = \frac{e^z}{z}. \tag{7}$$

Using the Taylor series of e^z, we get

$$f(z) = \frac{1}{z}\left(1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots\right). \tag{8}$$

Hence

$$\boxed{f(z) = \underbrace{\frac{1}{z}}_{\text{principal part}} + \underbrace{\left(1 + \frac{z}{2!} + \frac{z^2}{3!} + \cdots\right)}_{\text{regular part}}.} \tag{9}$$

The singularity is contained in the principal part; the regular part is still perfectly analytic at the origin.

Definition 3 (Residue)**.** If

$$f(z) = \sum_{n=-\infty}^{+\infty} c_n (z - z_0)^n, \tag{10}$$

then the residue of f at z_0 is the coefficient of $(z - z_0)^{-1}$:

$$\boxed{\operatorname{Res}(f, z_0) = c_{-1}.} \tag{11}$$

Example 2 (Why c_{-1} survives)**.** Let C_ρ be the positively oriented circle

$$z = z_0 + \rho e^{i\theta}, \qquad dz = i\rho e^{i\theta}\, d\theta, \qquad 0 \le \theta \le 2\pi. \tag{12}$$

For any integer n,

$$\oint_{C_\rho} (z - z_0)^n \, dz = i\rho^{n+1} \int_0^{2\pi} e^{i(n+1)\theta}\, d\theta. \tag{13}$$

If $n \neq -1$, the integral of the exponential over a full period vanishes. For $n = -1$,

$$\boxed{\oint_{C_\rho} \frac{dz}{z - z_0} = 2\pi i.} \tag{14}$$

Therefore, after integrating the Laurent series term by term, only the $(z - z_0)^{-1}$ term contributes:

$$\boxed{\oint_{C_\rho} f(z)\, dz = 2\pi i \ \mathrm{Res}(f, z_0).} \tag{15}$$

If z_0 is a pole of order m, the residue is computed by

$$\mathrm{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z \to z_0} \frac{d^{m-1}}{dz^{m-1}} \left[(z - z_0)^m f(z) \right]. \tag{16}$$

Example 3 (A pole of order two)**.** For

$$f(z) = \frac{e^z}{z^2}, \tag{17}$$

the Laurent expansion is

$$\frac{e^z}{z^2} = \frac{1}{z^2} + \frac{1}{z} + \frac{1}{2!} + \frac{z}{3!} + \cdots . \tag{18}$$

The residue is the coefficient of $1/z$, not the coefficient of $1/z^2$. The derivative formula gives

$$\mathrm{Res}\left(\frac{e^z}{z^2}, 0 \right) = \lim_{z \to 0} \frac{d}{dz} \left[z^2 \frac{e^z}{z^2} \right] = \lim_{z \to 0} e^z = 1. \tag{19}$$

For a simple pole, $m = 1$, no derivative is needed:

$$\boxed{\mathrm{Res}(f, z_0) = \lim_{z \to z_0} (z - z_0) f(z).} \tag{20}$$

4 Residues of the Model Function

For

$$f(z) = \frac{e^{-iaz}}{(z - b)(z + b)}, \tag{21}$$

the residue at $z = b$ is

$$\mathrm{Res}(f, b) = \lim_{z \to b} \frac{e^{-iaz}}{z + b} = \frac{e^{-iab}}{2b}. \tag{22}$$

At $z = -b$,

$$\mathrm{Res}(f, -b) = \lim_{z \to -b} \frac{e^{-iaz}}{z - b} = -\frac{e^{iab}}{2b}. \tag{23}$$

5 Detouring Around a Real Pole

Proposition 1 (Infinitesimal semicircle)**.** *Let z_0 be a simple pole of f, and let γ_ϵ be a semicircle of radius ϵ centered at z_0. Then*

$$\boxed{\lim_{\epsilon \to 0^+} \int_{\gamma_\epsilon} f(z)\, dz = \begin{cases} +i\pi \ \mathrm{Res}(f, z_0), & \textit{counterclockwise detour,} \\ -i\pi \ \mathrm{Res}(f, z_0), & \textit{clockwise detour.} \end{cases}} \tag{24}$$

Proof. Near z_0,

$$f(z) = \frac{\mathrm{Res}(f, z_0)}{z - z_0} + g(z), \tag{25}$$

where g is regular. Since g is bounded near z_0, and the length of a semicircle of radius ϵ is $\pi\epsilon$, the ML estimate gives

$$\left| \int_{\gamma_\epsilon} g(z)\, dz \right| \le M\pi\epsilon \to 0. \tag{26}$$

Thus only the singular term remains. For a counterclockwise upper semicircle,

$$z = z_0 + \epsilon e^{i\theta}, \qquad 0 \le \theta \le \pi, \tag{27}$$

and hence

$$\int_{\gamma_\epsilon} \frac{\mathrm{Res}(f, z_0)}{z - z_0}\, dz = i\pi \ \mathrm{Res}(f, z_0). \tag{28}$$

Reversing the orientation changes the sign. $\square$

6 Residues and Closed Contours

We now use the following elementary form of the residue theorem.

Proposition 2 (Contour without internal singularities)**.** *If f has no singularities inside a closed contour C, nor on C itself, then*

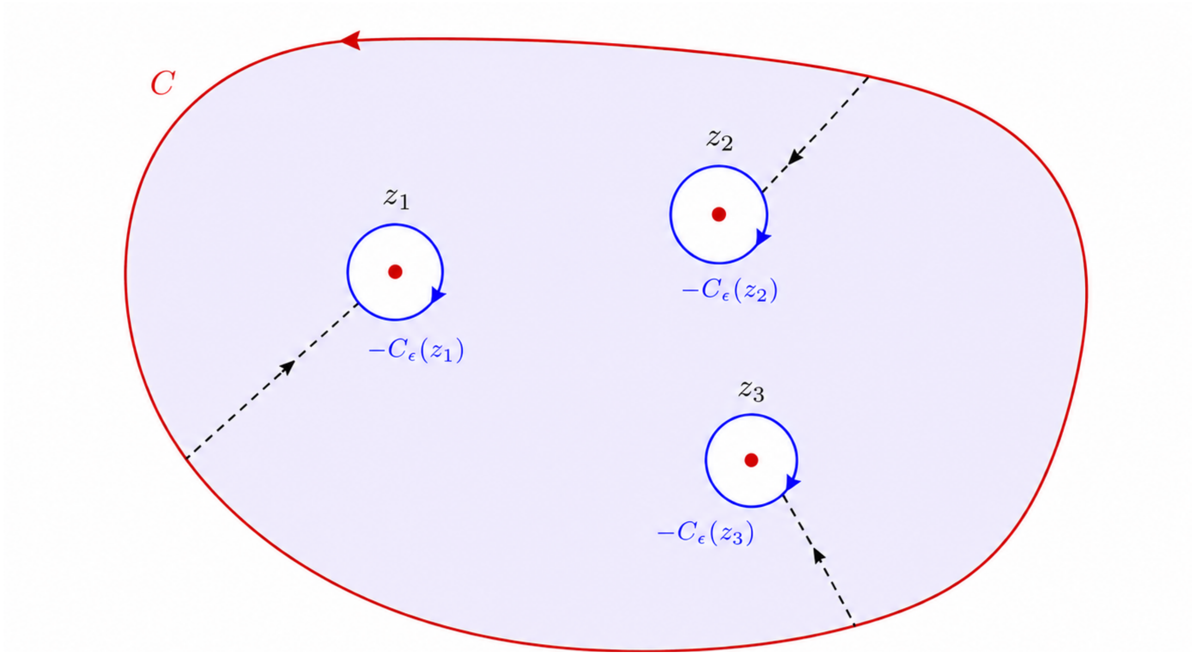
$$\boxed{\oint_C f(z)\, dz = 0.} \tag{29}$$

For a contour C enclosing simple poles $z_1, \ldots, z_N$, remove small disks $D_\epsilon(z_j)$ around the poles and define

$$D_\epsilon = \mathrm{Int}(C) \setminus \bigcup_{j=1}^{N} D_\epsilon(z_j). \tag{30}$$

The boundary of this perforated region consists of the outer contour C and the inner boundaries $-C_\epsilon(z_j)$, oriented clockwise because they are boundaries of holes:

$$\partial D_\epsilon = C \sqcup \big(-C_\epsilon(z_1)\big) \sqcup \cdots \sqcup \big(-C_\epsilon(z_N)\big). \tag{31}$$



$$D_\epsilon = \mathrm{Int}(C) \setminus \bigcup_{j=1}^{N} D_\epsilon(z_j)$$

Figure 1: Perforated region with dashed cuts. The outer contour C is counterclockwise. Each inner boundary $-C_\epsilon(z_j)$ is clockwise because it appears as the boundary of a hole. The cuts are traversed twice in opposite directions, so their contributions cancel.

Since f has no singularities in D_ϵ, the integral over the full boundary vanishes:

$$0 = \oint_{\partial D_\epsilon} f(z)\,dz = \oint_C f(z)\,dz - \sum_{j=1}^{N} \oint_{C_\epsilon(z_j)} f(z)\,dz. \tag{32}$$

Taking $\epsilon \to 0^+$, each small circle contributes $2\pi i\,\mathrm{Res}(f, z_j)$. Thus

$$\boxed{\oint_C f(z)\,dz = 2\pi i \sum_{j=1}^{N} \mathrm{Res}(f, z_j).} \tag{33}$$

7 Principal Value for $a < 0$

For $a < 0$, the factor e^{-iaz} decays on a large semicircle in the upper half-plane. We close the contour above and detour above the two real poles. The two small detours are clockwise, hence both have sign -1.

The large arc does not contribute. Indeed, for $z = Re^{i\theta}$, $0 \le \theta \le \pi$,

$$|e^{-iaz}| = e^{aR\sin\theta} \le 1 \qquad (a < 0), \tag{34}$$

and

$$|z^2 - b^2| \geq R^2 - b^2, \qquad \operatorname{long}(\Gamma_R) = \pi R. \tag{35}$$

Therefore

$$\left| \int_{\Gamma_R} f(z)\, dz \right| \leq \frac{\pi R}{R^2 - b^2} \xrightarrow[R\to\infty]{} 0. \tag{36}$$

Since the deformed contour contains no poles, the closed integral vanishes:

$$I + \int_{\gamma_{-b}} f(z)\, dz + \int_{\gamma_b} f(z)\, dz = 0. \tag{37}$$

Using the semicircle rule,

$$I = -i\pi \left[\operatorname{sgn}(\gamma_{-b}) \operatorname{Res}(f, -b) + \operatorname{sgn}(\gamma_b) \operatorname{Res}(f, b) \right]. \tag{38}$$

For the upper detours,

$$\operatorname{sgn}(\gamma_{-b}) = \operatorname{sgn}(\gamma_b) = -1. \tag{39}$$

Thus

$$I = \frac{i\pi}{2b} \left(e^{-iab} - e^{iab} \right) = \frac{\pi}{b} \sin(ab). \tag{40}$$

Therefore, for $a < 0$,

$$\boxed{\mathrm{PV} \int_{-\infty}^{+\infty} dx\, \frac{e^{-iax}}{x^2 - b^2} = \frac{\pi}{b} \sin(ab).} \tag{41}$$

8 Principal Value for $a > 0$

For $a > 0$, the large contour is closed in the lower half-plane, while the small detours around the real poles are kept above the poles. The large contour is now clockwise, so the closed contour integral is

$$\oint_{C_-} f(z)\, dz = -2\pi i \left[\operatorname{Res}(f, -b) + \operatorname{Res}(f, b) \right]. \tag{42}$$

The large lower arc again vanishes. The small detours are still clockwise, so

$$\int_{\gamma_{-b}} f(z)\, dz + \int_{\gamma_b} f(z)\, dz = -i\pi \left[\operatorname{Res}(f, -b) + \operatorname{Res}(f, b) \right]. \tag{43}$$

Thus

$$I = -i\pi \left[-\frac{e^{iab}}{2b} + \frac{e^{-iab}}{2b} \right] = -\frac{\pi}{b} \sin(ab). \tag{44}$$

Therefore, for $a > 0$,

$$\boxed{\mathrm{PV} \int_{-\infty}^{+\infty} dx\, \frac{e^{-iax}}{x^2 - b^2} = -\frac{\pi}{b} \sin(ab).} \tag{45}$$

The two cases combine into

$$\boxed{\mathrm{PV} \int_{-\infty}^{+\infty} dx\, \frac{e^{-iax}}{x^2 - b^2} = -\frac{\pi}{b} \operatorname{sgn}(a) \sin(ab), \qquad a \neq 0.} \tag{46}$$

9 The Feynman Prescription

The principal value keeps the poles on the real axis and interprets the integral through symmetric detours. The Feynman prescription does something different: it temporarily replaces the singular integrand by a regularized family whose poles are shifted away from the real axis.

Definition 4 (Feynman prescription)**.** The Feynman prescription moves the poles infinitesimally off the real axis, computes the regularized integral, and only at the end takes the limit $\epsilon \to 0^+$. In the present example,

$$b \mapsto b - i\epsilon, \qquad -b \mapsto -b + i\epsilon. \tag{47}$$

Thus the positive pole moves below the real axis and the negative pole moves above it.

We use

$$f_\epsilon(z) = \frac{e^{-iaz}}{(z - b + i\epsilon)(z + b - i\epsilon)}. \tag{48}$$

The poles are

$$z_+ = b - i\epsilon, \qquad z_- = -b + i\epsilon. \tag{49}$$

Expanding the denominator gives

$$(z - b + i\epsilon)(z + b - i\epsilon) = z^2 - b^2 + 2ib\epsilon + \epsilon^2. \tag{50}$$

Since ϵ^2 is neglected,

$$\boxed{\frac{1}{z^2 - b^2} \longrightarrow \frac{1}{z^2 - b^2 + i\,2b\epsilon}.} \tag{51}$$

The factor $2b$ is not lost; it can later be absorbed into a redefinition of the infinitesimal.

The residues of the regularized function are

$$\operatorname{Res}(f_\epsilon, z_-) = \frac{e^{-ia(-b+i\epsilon)}}{(-b + i\epsilon) - (b - i\epsilon)} \xrightarrow[\epsilon \to 0^+]{} -\frac{e^{iab}}{2b}, \tag{52}$$

and

$$\operatorname{Res}(f_\epsilon, z_+) = \frac{e^{-ia(b-i\epsilon)}}{(b - i\epsilon) - (-b + i\epsilon)} \xrightarrow[\epsilon \to 0^+]{} \frac{e^{-iab}}{2b}. \tag{53}$$

Case $a < 0$

For $a < 0$, close the contour in the upper half-plane. Only $z_- = -b + i\epsilon$ lies inside, and the orientation is positive:

$$I_F(a < 0) = 2\pi i \left(-\frac{e^{iab}}{2b} \right) = -\frac{\pi i}{b} e^{iab}. \tag{54}$$

Case $a > 0$

For $a > 0$, close the contour in the lower half-plane. Only $z_+ = b - i\epsilon$ lies inside, but the contour is clockwise:

$$I_F(a > 0) = -2\pi i \left(\frac{e^{-iab}}{2b} \right) = -\frac{\pi i}{b} e^{-iab}. \tag{55}$$

Both cases are summarized by

$$\boxed{I_F(a) = -\frac{\pi i}{b} e^{-i|a|b}, \qquad a \neq 0.} \tag{56}$$

Its real part reproduces the principal value:

$$\boxed{\operatorname{Re} I_F(a) = -\frac{\pi}{b} \operatorname{sgn}(a) \sin(ab).} \tag{57}$$

The imaginary part,

$$\operatorname{Im} I_F(a) = -\frac{\pi}{b} \cos(ab), \tag{58}$$

is the extra information carried by the Feynman prescription.

10 Textbook Notation

Since $b > 0$ is finite, we may define

$$\eta = 2b\epsilon, \qquad \eta \to 0^+. \tag{59}$$

Then

$$z^2 - b^2 + i\,2b\epsilon = z^2 - b^2 + i\eta. \tag{60}$$

Renaming η as ϵ, one obtains the usual notation

$$\boxed{\frac{1}{z^2 - b^2} \longrightarrow \frac{1}{z^2 - b^2 + i\epsilon}, \qquad \epsilon \to 0^+.} \tag{61}$$

With this convention, the final result is

$$\boxed{\int_{-\infty}^{+\infty} dx\, \frac{e^{-iax}}{x^2 - b^2 + i\epsilon} = -\frac{\pi i}{b} e^{-ib|a|}, \qquad a \neq 0.} \tag{62}$$

Absorbing the finite factor before computing the poles gives the same answer. Indeed,

$$z^2 - b^2 + i\eta = 0 \qquad \Longrightarrow \qquad z = \pm\sqrt{b^2 - i\eta}. \tag{63}$$

For $\eta \ll 1$,

$$\sqrt{b^2 - i\eta} = b\sqrt{1 - \frac{i\eta}{b^2}} \simeq b - \frac{i\eta}{2b}. \tag{64}$$

Thus

$$z_+ = b - \frac{i\eta}{2b}, \qquad z_- = -b + \frac{i\eta}{2b}. \tag{65}$$

If $\eta = 2b\epsilon$, these are precisely

$$z_+ = b - i\epsilon, \qquad z_- = -b + i\epsilon. \tag{66}$$

11 Connection with the Feynman Propagator

The role of the $+i\epsilon$ term is now clear: it moves the poles off the real axis and fixes the contour prescription. It is not a finite physical correction to the denominator; it is an instruction for how the integral is to be understood.

In quantum field theory, the scalar Feynman propagator is written as

$$\boxed{\Delta_F(x-y) = i\int \frac{d^4k}{(2\pi)^4}\frac{e^{-ik\cdot(x-y)}}{k^2-m^2+i\epsilon}.} \tag{67}$$

The denominator $k^2-m^2+i\epsilon$ must be read with the same logic as the one-dimensional example: the $+i\epsilon$ tells us how the poles in the complex k^0-plane are displaced. That choice defines the integration contour and selects the Feynman propagator.